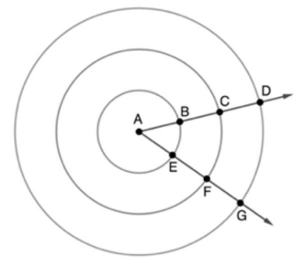
Geometry
Unit 10
Lesson 3 Homework

Name___Answer Key_______
Date____________________
Period___________________

For questions 1-6, concentric circles with center A are shown. Points B and E lay on the innermost Circle A, and has radius $AB = 2$ units. Points C and F lay on the middle Circle A, and has radius $AC = 4$ units. Points D and G lay on the outer Circle A, and has radius $AD = 6$ units.



Determine the circumference for each circle. Leave your answer in terms of π.

	Circle	Circumference
1.	Innermost Circle	4π
2.	Middle Circle	8π
3.	Outer Circle	12π

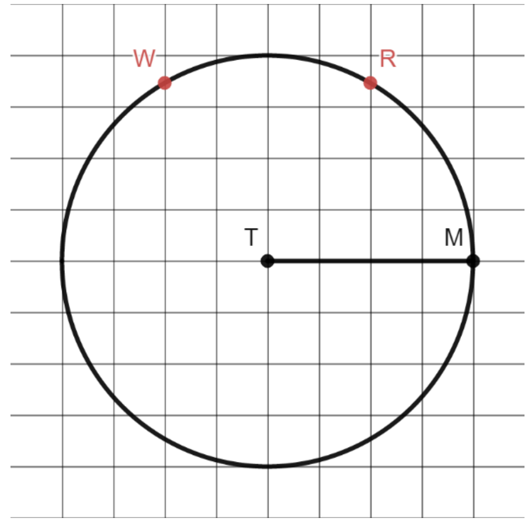
Complete the table.

	Circle	Ratio of Circumference to Radius
4.	Innermost Circle	$\frac{4\pi}{2} = 2\pi$
5.	Middle Circle	$\frac{8\pi}{4} = 2\pi$
6.	Outer Circle	$\frac{12\pi}{6} = 2\pi$

7. What would be the ratio of the circumference to the radius for a circle whose center is N and whose radius has a measurement of 9?

$\frac{18\pi}{9} = 2\pi$

Circle T is shown where $\overline{TM}$ is a radius.



8. Label point R on the circumference of the circle so that the resulting arc has an approximate length of 1 radian.
See grid.

9. Describe how you determined the location of point R.
Answers will vary. Example: Measured the length of the radius and measured the same length on the circle.

10. Label point W on the circumference of the circle so that the resulting arc has an approximate length of 2 radians.
See grid.